WoT and the Agentic Web

Ege Korkan, Siemens

W3C TPAC 2026, Dublin

Unrestricted | Ege Korkan | © Siemens 2026

SIEMENS

Depending on the Audience

- Basic WoT introduction
- Planned deliverables in new charter

SIEMENS

Agentic AI with WoT

Mostly a discussion, early stage

SIEMENS

Multiple Trends?

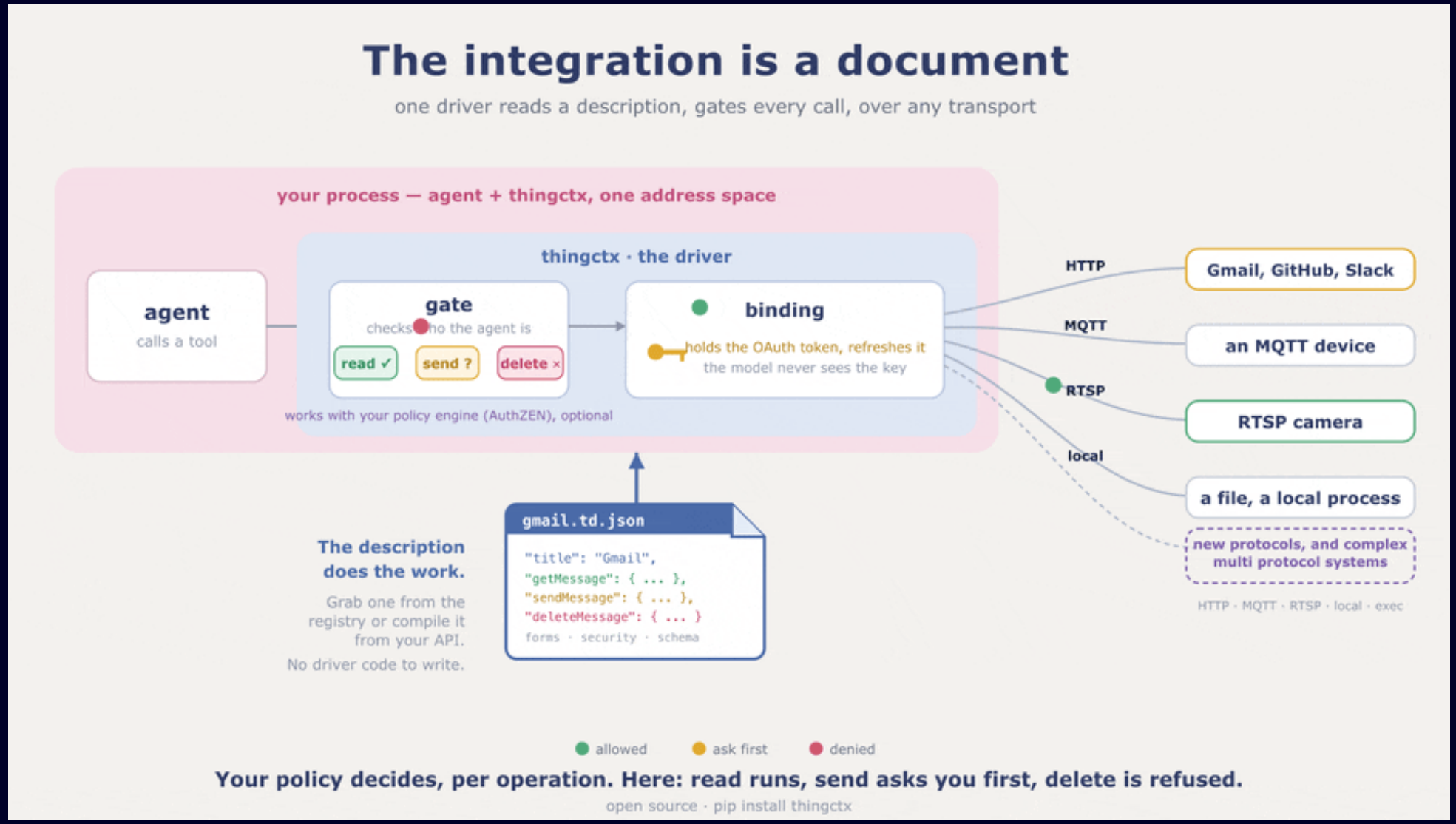
- Physical AI: Simulations but also interaction with the physical world
  - Interaction with non-communicating, communicating and passive, and proactive entities
- ”IoT protocol fight” is stabilizing and value needs to be more visible in a scalable way
  - PoCs and demos are not convincing anyone. Stability, open standards, semantics, interoperability
- Agentic AI is more than a trend
  - Letting one or more agent do their tasks. What if they need to automate the physical world?
- New standards are emerging to bring agents to the physical world and to coordinate them
  - Model Hardware Standard, A2A, and more
- Web of Things Working Group is rechartering
  - https://www.w3.org/2026/09/wot-wg-2026.html
  - **WoT Primer and Best Practice:** Document supporting developers in designing WoT-based applications, including the combination of AI techniques and tooling (e.g., AI Agents and MCP).

SIEMENS

Some are already using WoT in agentic systems

SIEMENS

Thingctx



https://github.com/thingctx/thingctx

SIEMENS

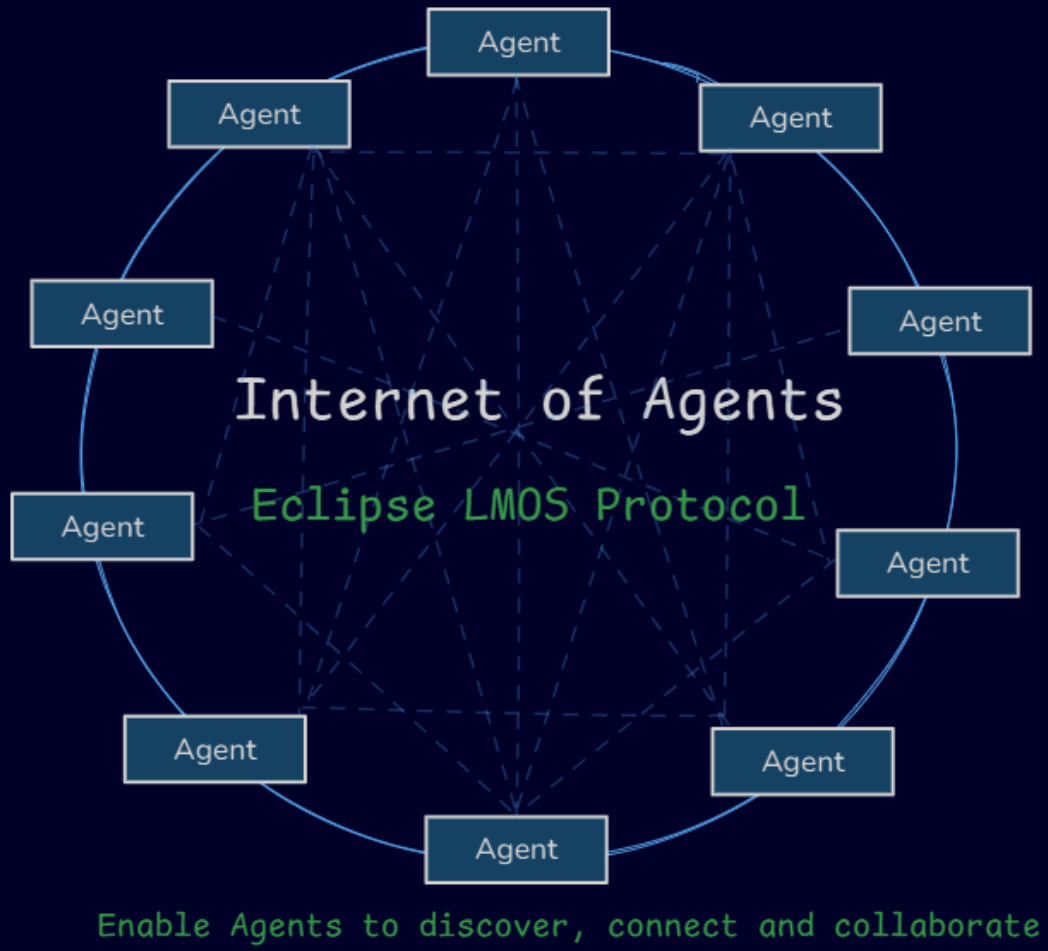
thingctx



https://www.youtube.com/watch?v=3QdxP7ydAB0

SIEMENS

Eclipse LMOS (https://eclipse.dev/lmos/)



The LMOS Agent Description Format builds on top of the Thing Description (TD) format from the Web of Things (WoT) specification and extends it by specifying an additional schema tailored to the needs of intelligent agents.

Unrestricted | Ege Korkan | © Siemens 2026

SIEMENS

Basic Web of Things Introduction

All information is provided to have a reference. Try the tutorial!

https://w3c-cg.github.io/wot-cg/tutorials/whatiswot/docs/intro.html

SIEMENS

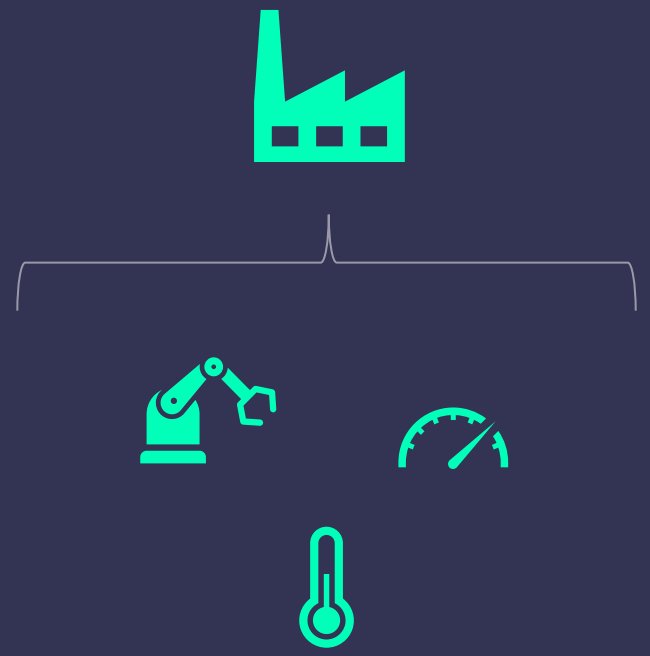
You can get hands-on

- Room 5 on the second floor
- Full day today and tomorrow, then demo as a breakout on Wednesday’s final slot

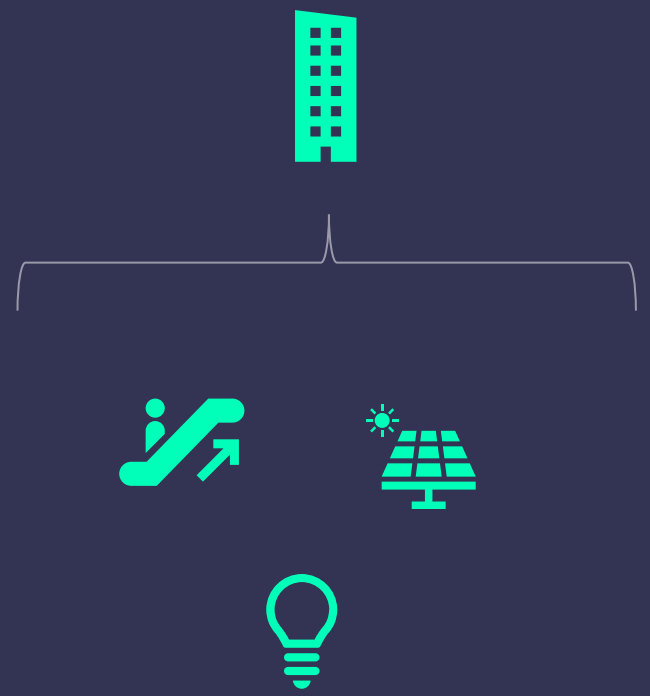
SIEMENS

It’s about the *Things* that are used in application domains

Factory



Building



City



...

SIEMENS

There is a common problem if you want to use Things smart: Onboarding

Monitoring

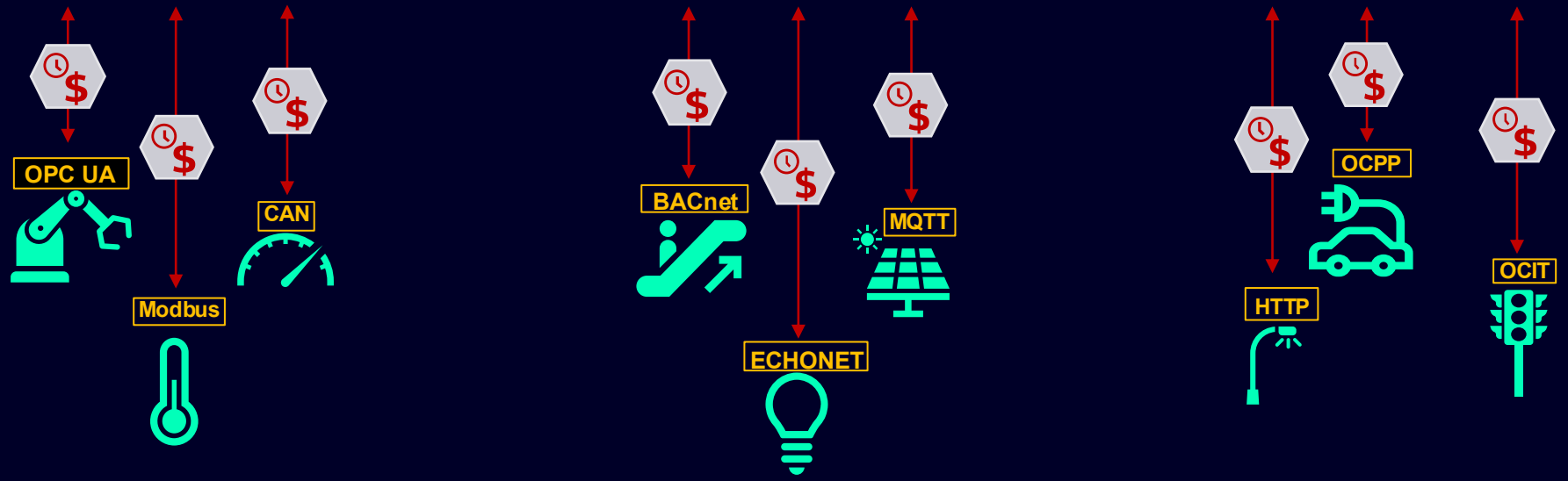
Controlling

Cloud Services

AI

...

That's what customers want



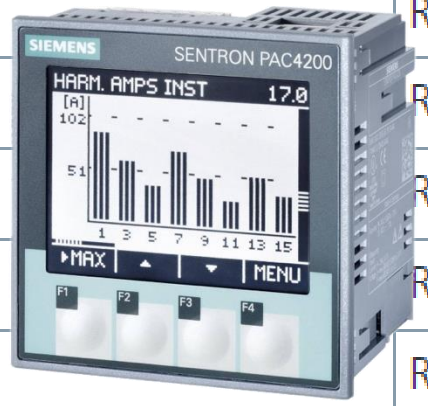
But the real effort is under the hood:

IoT asset's interface based on their specific protocols / functions, data models and security mechanism

Time & cost intensive integration + maintenance

SIEMENS

57	2	3-phase average voltage L-N	Float	V	-						R
59	2	3-phase average voltage L-L	Float	V	-						R
61	2	3-Phase Average Current L-L	Float	A	-						R
63	2	Total Apparent Power	Float	VA	-						R
65	2	Total Active Power	Float	W	-						R
67	2	Total Reactive Power (Qn)	Float	VAR	-						R

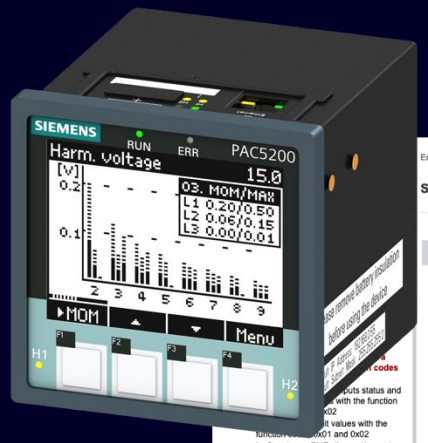


R	32792	8018	2	var	1	R	Fl32	Total reactive power vectorial	Vectorial total reactive power according to DIN EN 61557-12 with sign as sum of the individual reactive powers. >0: Demand
R	32794	801A	2	VA	1	R	Fl32	Total apparent power vectorial	Vectorial tota ccording
R	32796	801C	2	-	1	R	Fl32	Total power factor vectorial	Vectorial to sign according >0: Inducti <0: Capacitive
R	32798	801E	2	W	1	R	Fl32	Active power phase 1	Active power with sign on phase 1. >0: Demand <0: Deliver



Image source: Siemens and PhoenixContact

Manually transformation process Examples



Entry type: **Manual**, Entry ID: **34261595**, Entry date: **05/27/2019**

System Manual SENTRON PAC4200 Power Monitoring Device

Document: PAC4200 (05/2019, English)
Type of topic: Fact

Measured variables without a time stamp with the function codes 0x03 and 0x04

Addressing the measured variables without a time stamp

The SENTRON PAC4200 Power Monitoring Device provides measured variables with or without a time stamp.

Note

Error in the case of inconsistent access to measured values

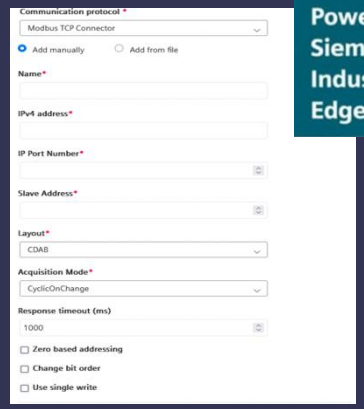
Please ensure the start offset of the register is correct when making **read accesses**.

Please ensure the start offset and the number of registers are correct when making **write accesses**.

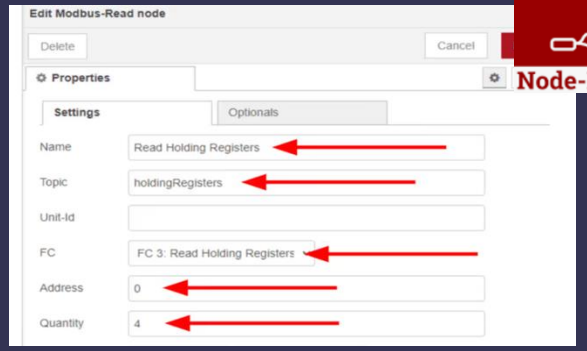
If a value consists of two registers, a read command applied in the second register, for example, will generate an error code. The SENTRON PAC4200 will also output an error code if, for example, a write operation ends in the middle of a multi-register value.

Table: Measured variables available without a time stamp

Offset	Number of registers	Name	Format	Unit	Value range	Access
1	2	Voltage L1-N	Float	V	-	R
3	2	Voltage L2-N	Float	V	-	R
5	2	Voltage L3-N	Float	V	-	R
7	2	Voltage L1-L2	Float	V	-	R
9	2	Voltage L2-L3	Float	V	-	R



Powered by Siemens Industrial Edge



Node-RED

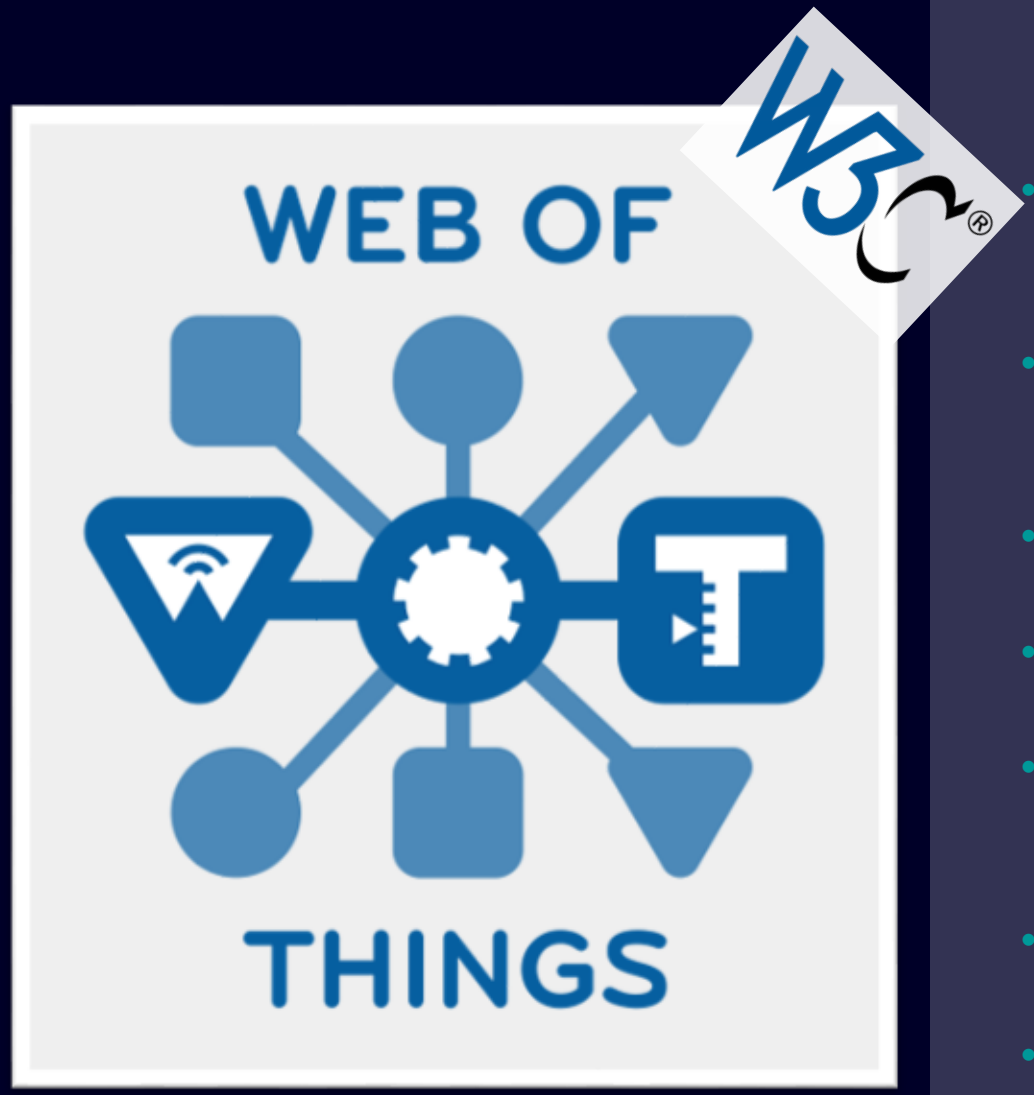
```
<tag>
    <tagDefinition>
        <unitId>0</unitId>
        <startIdx>11</startIdx>
        <readType>HOLDING_REGISTERS</readType>
        <dataType>INT_64</dataType>
    </tagDefinition>
    <tagName>myTag</tagName>
</tag>
```

HIVEMQ EDGE

```
modbus:
  - name: hub1
    type: tcp
    host: IP_ADDRESS
    port: 502
    binary_sensors:
      - name: my_relay
        address: 100
        device_class: door
        input_type: coil
        scan_interval: 15
```


SIEMENS

Enter the W3C Web of Things



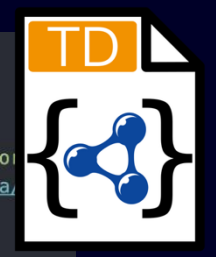
- Based on IT-Friendly technologies, reuse established Web standards
- Works with all protocols (MQTT, HTTP, Modbus, OPC UA, BACnet, KNX IoT,…)
- Adapts to any application domain
- Local, Edge, Cloud
- Key technology: Standardized Device Description Language called **WoT Thing Description (TD)**
- Open and royalty-free W3C standard
- Published the 1.1 version; currently working on WoT 2.0

Unrestricted | Ege Korkan | © Siemens 2026

WoT Thing Description

Standardized machine and human readable device interface descriptions

```jsonc
{
    "@context": "https://www.w3.org/2022/wot/td/v1.1",
    "title": "Siemens SENTRON PAC4200",
    "base": "modbus+tcp://192.168.10.100:502/1/",
    "description": "The SENTRON PAC4200 is a measuring device fo
    "support": "https://support.industry.siemens.com/dl/dl-media/
    "securityDefinitions": {
        "nosec_sc": {
            "scheme": "nosec"
        }
    },
    "security": "nosec_sc",
    "properties": {
        "voltage_l1_n": {
            "title": "Voltage L1-N",
            "type": "number",
            "unit": "V",
            "forms": [
                {
                    "href": "40001?quantity=2",
                    "contentType": "application/octet-stream",
                    "modv:function": "readHoldingRegisters",
                    "modv:type": "xsd:float",
                    "modv:mostSignificantByte" : true,
                    "modv:mostSignificantWord" : true,
                }
            ]
        },
        "voltage_l2_n": {
            "title": "Voltage L2-N",
            "type": "number",
            "unit": "V",
            "forms": [
                {
                    "href": "40003?quantity=2",
                    "contentType": "application/octet-stream",
```





W3C Recommendation

Web of Things (WoT) Thing Description

W3C Recommendation 9 April 2020 (Link errors corrected 23 June 2020)

This version:
https://www.w3.org/TR/2020/REC-wot-thing-description-20200409/

Latest published version:
https://www.w3.org/TR/wot-thing-description/

Latest editor's draft:
https://w3c.github.io/wot-thing-description/

Implementation report:
https://w3c.github.io/wot-thing-description/testing/report.html

Previous version:
https://www.w3.org/TR/2020/PR-wot-thing-description-20200130/

Editors:
Sebastian Kaebisch (Siemens AG)
Takuki Kamiya (Fujitsu Laboratories of America)
Michael McCool (Intel)
Victor Charpenay (Siemens AG)
Matthias Kovatsch (Huawei)

Participate:
GitHub w3c/wot-thing-description
File a bug
Commit history
Pull requests

Contributors:
In the GitHub repository

Repository:
We are on GitHub
File a bug

Please check the **errata** for any errors or issues reported since publication.

See also **translations**.

Copyright © 2017-2020 W3C® (MIT, ERCIM, Keio, Beihang). W3C liability, trademark and permissive document license rules apply.

Abstract

WoT Thing Description

Standardized machine and human readable device interface descriptions




```
{
    "@context": "https://www.w3.org/2022/wot/td/v1.1",
    "title": "Siemens SENTRON PAC4200",
    "base": "modbus+tcp://192.168.10.100:502/1/",
    "description": "The SENTRON PAC4200 is a measuring device fo
    "support": "https://support.industry.siemens.com/dl/dl-media/
    "securityDefinitions": {
        "nosec_sc": {
            "scheme": "nosec"
        }
    },
    "security": "nosec_sc",
    "properties": {
        "voltage_l1_n": {
            "title": "Voltage L1-N",
            "type": "number",
            "unit": "V",
            "forms": [
                {
                    "href": "40001?quantity=2",
                    "contentType": "application/octet-stream",
                    "modv:function": "readHoldingRegisters",
                    "modv:type": "xsd:float",
                    "modv:mostSignificantByte" : true,
                    "modv:mostSignificantWord" : true,
                }
            ]
        },
        "voltage_l2_n": {
            "title": "Voltage L2-N",
            "type": "number",
            "unit": "V",
            "forms": [
                {
                    "href": "40003?quantity=2",
                    "contentType": "application/octet-stream",
```



Provides details about …

… the interface endpoint
(e.g., IP and port address)

… the used protocols and serializations
(e.g., Modbus, OPC UA, UA Binary, JSON, …)

… what kind of data and functions are served
(e.g., voltage values, on/off,…)

… how the payload structure look like
(e.g., simple types, complex types)

… context and semantics
(e.g., based on ECLASS, OPC UA DM, …)

… security requirements
(e.g., Bearer, OAuth2, …)

Unrestricted | Ege Korkan | © Siemens 2026

WoT Bindings
Native integration into diverse ecosystems

W3C Editor's Draft

Web of Things (WoT) HTTP Binding

W3C Editor's Draft 22 April 2026

▾ More details about this document

This version:
https://w3c.github.io/wot-binding-templates/bindings/protocols/http/

Latest published version:
https://www.w3.org/TR/wot-http-binding/

Latest editor's draft:
https://w3c.github.io/wot-binding-templates/bindings/protocols/http/

History:
Commit history

Editors:
Michael Koster (Invited Expert)
Ege Korkan (Siemens AG)

Feedback:
GitHub w3c/wot-binding-templates (pull requests, new issue, open issues)

W3C Editor's Draft

Web of Things (WoT) CoAP Binding

W3C Editor's Draft 22 April 2026

▾ More details about this document

This version:
https://w3c.github.io/wot-binding-templates/bindings/protocols/coap/

Latest published version:
https://www.w3.org/TR/wot-coap-binding/

Latest editor's draft:
https://w3c.github.io/wot-binding-templates/bindings/protocols/coap/

History:
Commit history

Editors:
Klaus Hartke (Siemens AG)
Jan Romann (Invited Expert)

Feedback:
GitHub w3c/wot-binding-templates (pull requests, new issue, open issues)

W3C Editor's Draft

Web of Things (WoT) PROFINET Binding

W3C Editor's Draft 22 April 2026

▾ More details about this document

This version:
https://w3c.github.io/wot-binding-templates/bindings/protocols/profinet/

Latest published version:
https://www.w3.org/TR/wot-profinet-binding/

Latest editor's draft:
https://w3c.github.io/wot-binding-templates/bindings/protocols/profinet/

History:
Commit history

Editor:
Kazeem Oladipupo (Siemens AG)

Feedback:
GitHub w3c/wot-binding-templates (pull requests, new issue, open issues)

W3C Editor's Draft

Web of Things (WoT) MQTT Binding

W3C Editor's Draft 22 April 2026

▾ More details about this document

This version:
https://w3c.github.io/wot-binding-templates/bindings/protocols/mqtt/

Latest published version:
https://www.w3.org/TR/wot-mqtt-binding/

Latest editor's draft:
https://w3c.github.io/wot-binding-templates/bindings/protocols/mqtt/

History:
Commit history

Editors:
Michael Koster (Invited Expert)
Ege Korkan (Siemens AG)
Cristiano Aguzzi (Invited Expert)

Feedback:
GitHub w3c/wot-binding-templates (pull requests, new issue, open issues)

W3C Editor's Draft

Web of Things (WoT) Modbus Binding

W3C Editor's Draft 22 April 2026

▾ More details about this document

This version:
https://w3c.github.io/wot-binding-templates/bindings/protocols/modbus/

Latest published version:
https://www.w3.org/TR/wot-modbus-binding/

Latest editor's draft:
https://w3c.github.io/wot-binding-templates/bindings/protocols/modbus/

History:
Commit history

Editor:
Cristiano Aguzzi (Invited Expert)

Feedback:
GitHub w3c/wot-binding-templates (pull requests, new issue, open issues)

W3C Editor's Draft

Web of Things (WoT) BACnet Binding

W3C Editor's Draft 22 April 2026

▾ More details about this document

This version:
https://w3c.github.io/wot-binding-templates/bindings/protocols/bacnet/

Latest published version:
https://www.w3.org/TR/wot-bacnet-binding/

Latest editor's draft:
https://w3c.github.io/wot-binding-templates/bindings/protocols/bacnet/

History:
Commit history

Editors:
Klaus Hartke (Siemens AG)
Michael Koster (Invited Expert)
Dogan Fennibay (Siemens AG)

Feedback:
GitHub w3c/wot-binding-templates (pull requests, new issue, open issues)

W3C Editor's Draft

Web of Things (WoT) LoRaWAN Binding

W3C Editor's Draft 11 May 2026

▾ More details about this document

This version:
https://w3c.github.io/wot-binding-templates/bindings/protocols/lorawan

Latest published version:
https://www.w3.org/TR/wot-lorawan-binding/

Latest editor's draft:
https://w3c.github.io/wot-binding-templates/bindings/protocols/lorawan

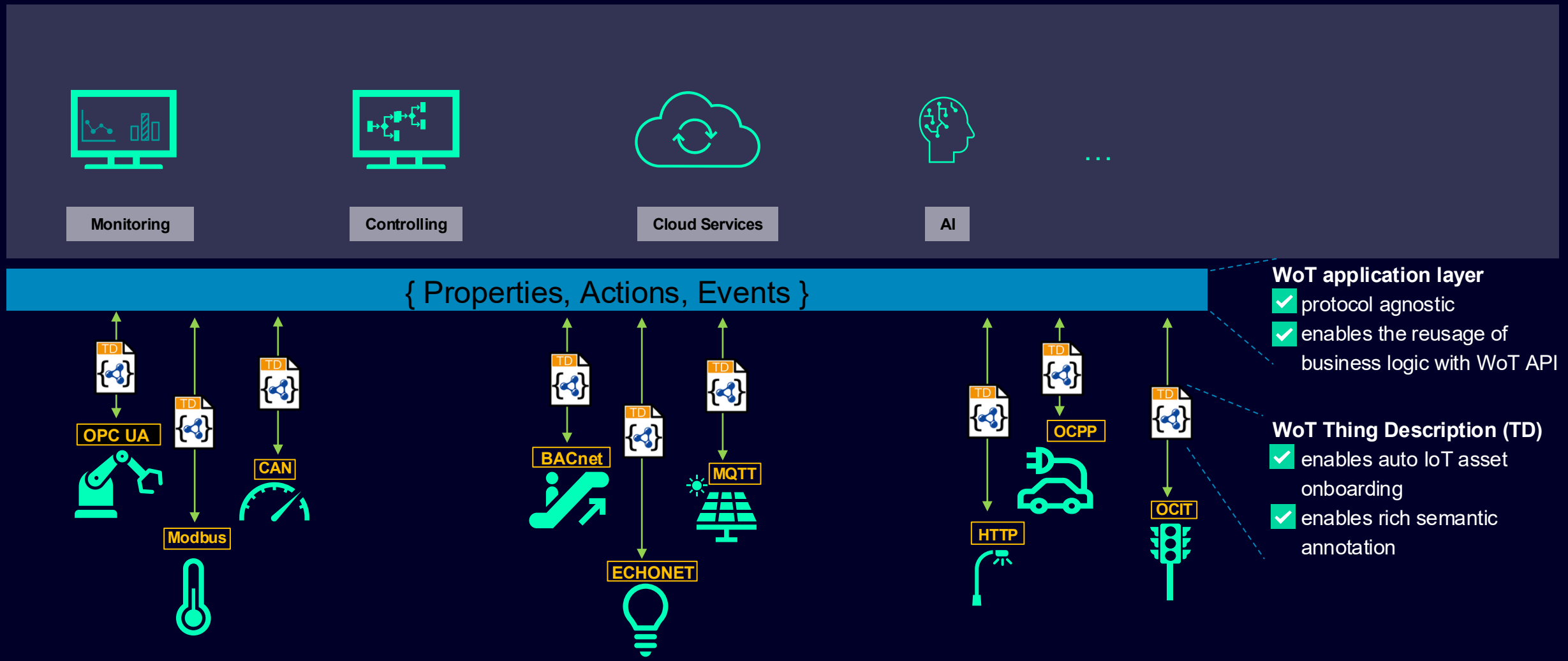
History:
Commit history

Editor:
Erich Barnstedt

Feedback:
GitHub w3c/wot-binding-templates (pull requests, new issue, open issues)

SIEMENS

W3C WoT simplifies the Thing onboarding and IoT application development



WoT application layer

- protocol agnostic
- enables the reusage of business logic with WoT API

WoT Thing Description (TD)

- enables auto IoT asset onboarding
- enables rich semantic annotation

Unrestricted | Ege Korkan | © Siemens 2026

SIEMENS

Deeper look into the Thing Description

W3C Recommendation

TABLE OF CONTENTS

Abstract

Status of This Document

1. Introduction
1.1 Thing Description
1.2 Thing Model

2. Conformance

3. Terminology

4. Namespaces

5. TD Information Model
5.1 Overview
5.2 Preliminaries
5.3 Class Definitions
5.3.1 Core Vocabulary Definitions
5.3.1.1 `Thing`
5.3.1.2 `InteractionAffordance`
5.3.1.3 `PropertyAffordance`
5.3.1.4 `ActionAffordance`
5.3.1.5 `EventAffordance`
5.3.1.6 `VersionInfo`
5.3.1.7 `MultiLanguage`
5.3.2 Data Schema Vocabulary Definitions
5.3.2.1 `DataSchema`

Web of Things (WoT) Thing Description 1.1

W3C Recommendation 05 December 2023

▼ **More details about this document**

This version:
https://www.w3.org/TR/2023/REC-wot-thing-description11-20231205/

Latest published version:
https://www.w3.org/TR/wot-thing-description11/

Latest editor's draft:
https://w3c.github.io/wot-thing-description/

History:
https://www.w3.org/standards/history/wot-thing-description11/
Commit history

Implementation report:
https://w3c.github.io/wot-thing-description/testing/report11.html

Editors:
Sebastian Kaebisch (Siemens AG)
Michael McCool (Intel Corp.)
Ege Korkan (Siemens AG)

Former editors:
Takuki Kamiya (Fujitsu Research of America)
Victor Charpenay (when at Siemens AG)
Matthias Kovatsch (when at Huawei)

Feedback:

SIEMENS

Deeper look into the Thing Description: **Thing Level**

```
{
        "@context": [
                "https://www.w3.org/2022/wot/td/v11",
                {"iot":"http://iotschema.org"}
        ],
        "@type":"SprinklerStation",
        "id": "urn:dev:ops:Agriculture-7331",
        "title": "SoilStation",
        "description": "A soil management station for use in agriculture",
        "securityDefinitions": { "basic_sc": {"scheme": "basic"} },
        "security": "basic_sc",
        "properties": {
                "temperature": {...}
        },
        "actions": {
                "startSprinkler": {...}
        },
        "events": {
                "tooDry": {...}
        },
        "links":["href":"docs", "rel":"about", "type":"text/html"]
}
```

Thing Metadata

Semantic Annotations

Security Metadata

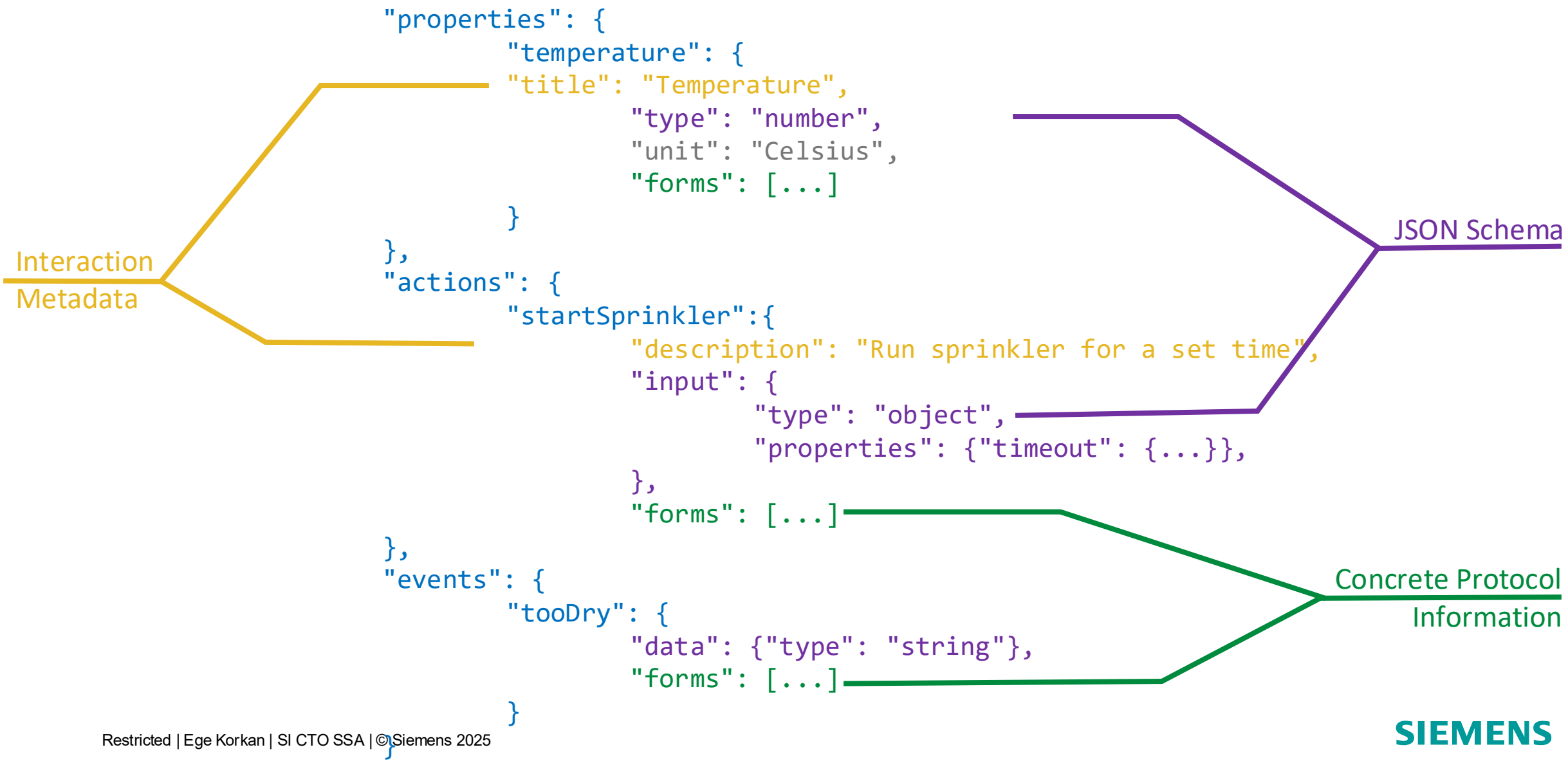
Definitions of Interaction Affordances

Links to other Documents

Restricted | Ege Korkan | SI CTO SSA | © Siemens 2025

SIEMENS

Deeper look into the Thing Description: **Interaction Level**



SIEMENS

Deeper look into the Thing Description: **Protocol Level**



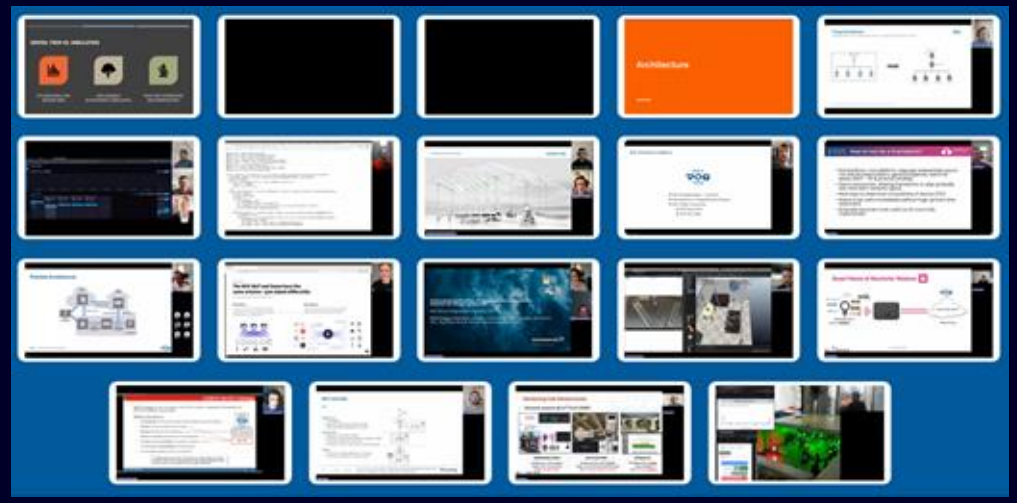
Restricted | Ege Korkan | SI CTO SSA | © Siemens 2025

SIEMENS

External Adoption

SIEMENS

External Adoption



34 Meetups and counting



WoT@Industry 2024 in Munich

SIEMENS

Where to learn about external adoption?

- Community Group:
  - Meetups are online events where adopters can show their project, product or research.
  - Videos are available on YouTube
  - Slides and meeting minutes are available on GitHub
  - Alternative link from Internal Sharepoint: Web of Things Meetups
- WoT Homepage: https://www.w3.org/WoT

Restricted | © Siemens 2026 | Ege Korkan | SI CTO SSA

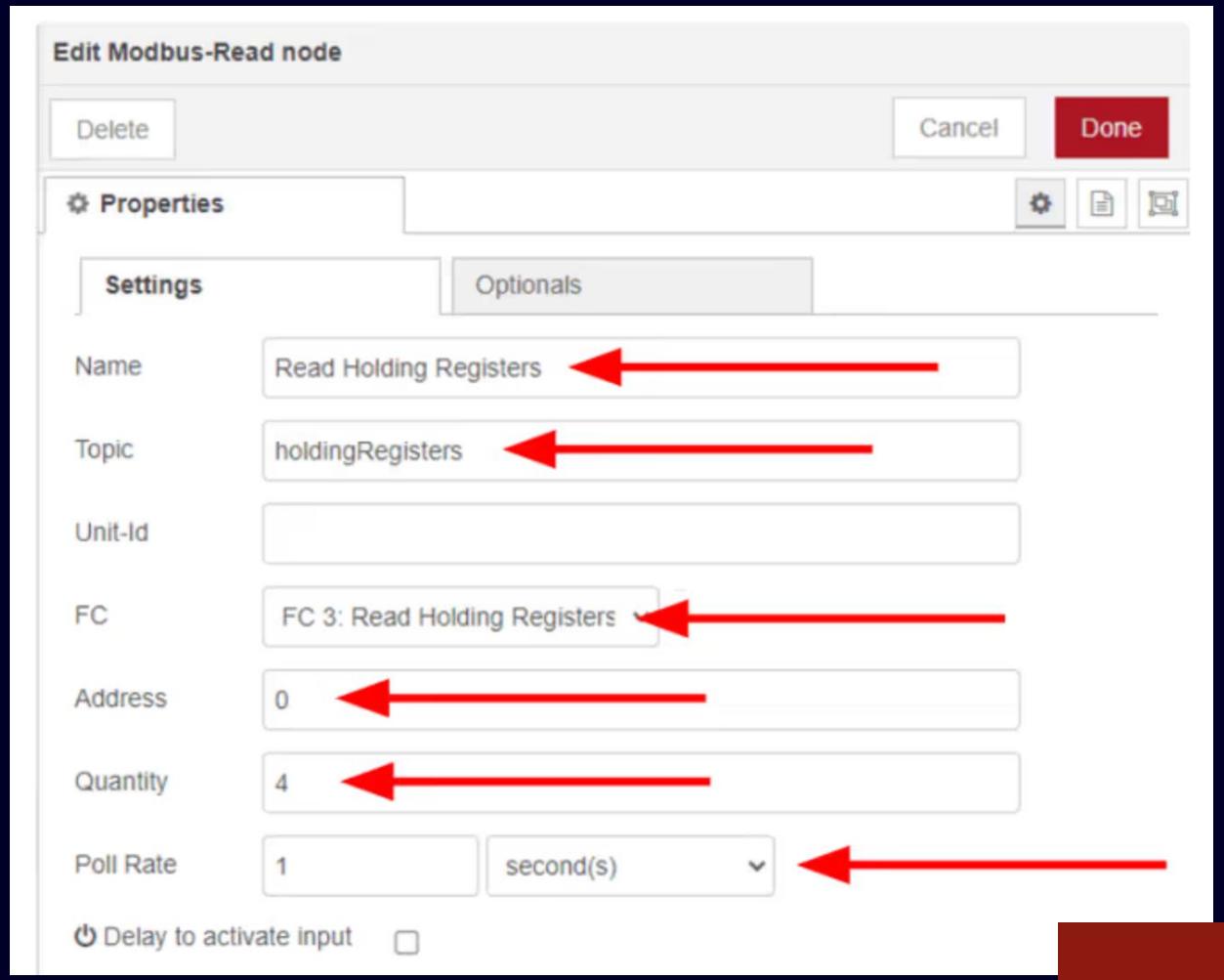
Device Catalog

Not writing JSONs

Unrestricted | Ege Korkan | © Siemens 2026

SIEMENS

Example TD for the Node-RED example from before



Node-RED

TD Extract

does not repeat

```
"base":"modbus+tcp://192.168.0.10:502",
...
{
  "href": "0?quantity=4",
  "modv:function":"readHoldingRegisters",
  "modv:type":"xsd:float",
  "modv:mostSignificantByte": true,
  "modv:mostSignificantWord": true
}
```

repeats for each instance

Unrestricted | Ege Korkan | © Siemens 2026

SIEMENS



Thing Model catalog

Tooling, home and services

https://github.com/wot-oss/tmc

SIEMENS

Thing Model Catalog: CLI

README Apache-2.0 license

Thing Model Catalog CLI

go report A+ release v0.1.2 go.dev docs license scan passing



Thing Model catalog

Find, use and contribute device descriptions for industrial IoT devices!

```
Usage:
  tmc [command]

Available Commands:
  attachment  Manage TM attachments
  check       Check the integrity of all or only named resources in repository's inte
  completion  Generate the autocompletion script for the specified shell
  copy        Copy multiple TMs and their attachments from one repository to another
  create-si   Create or update search index
  delete      Delete a TM by id
  export      Export multiple TMs from a catalog and, optionally, their attachments
  fetch       Fetch a TM by name or id
  help        Help about any command
  import      Import a TM or a directory with TMs into a catalog
  index       Refresh the repository's internal index, if it has one
  list        List TMs in catalog
  repo        Manage repositories
  search      Search full text of TMs in catalog using bleve search engine
  serve       Start a REST API server
  validate    Validate a TM before importing
  version     Show tmc version information
  versions    List available versions of the TM with given name
```

https://www.github.com/wot-oss/tmc

Also available with a REST API

Restricted | © Siemens 2026 | Ege Korkan | SI CTO SSA

SIEMENS

Going further than Onboarding

On your own, with the community, commercially

SIEMENS

Developer Resources

There are various resources available to build Web of Things applications. These are libraries, ready-to-use software, services or SDKs, which can be used in different stages of development or for development needs and are grouped below.

Category All | Thing Description | WoT Development | Others

Platform All | Application | Browser | CLI | IDE Extension | Library | Service

Language All | C# | Dart | Go | Java | Kotlin | Python | Rust | TypeScript

Obsolete* Show | Hide

Showing 41 matching tools

Editing and Validation

Tools for editing, validating, and parsing Thing Descriptions in various programming environments

Thing Description Playground	Browser application for validating and designing W3C Thing Descriptions and Thing Models.
Eclipse edi{TD}or	Eclipse ediTDor - A tool for simply designing W3C Thing Descriptions
TD-code	VS Code extension for validating and designing W3C Thing Descriptions and Thing Models.
Eclipse Ditto :: WoT :: Model	This module contains a W3C WoT (Web of Things) Java model following the Web of Things (WoT) Thing Description 1.1
Java API for Thing Descriptions of WoT (JDTs)	Java API for Thing Descriptions of the Web of Things

Build Agentic Systems

Java Node.js Kotlin

Program your Own Application

Auto-generate UIs

Dart Python Go .NET

Auto-generate Node-RED Flows

Auto-test Devices' Endpoints

Transform TDs to Other Formats

Host TDs with a Standardized API

Build Intermediaries

SIEMENS

Simple Scripting Example: Read Sensor Values and publish to MQTT Broker

```
const client = mqtt.connect("mqtt://test.mosquitto.org");
const baseTopic = “buildings/site1/heater/”

const sentronTD = WoT.fetch(TD_LOCATION)
let sentronThing = WoT.consume(sentronTD)

client.on("connect", () => {

 setInterval(()=>{

   const curVoltage1 =  sentronThing.readProperty("VoltageL1-N")

   client.publish(baseTopic+"voltagel1n", curVoltage1);

 }, 10000) // read and publish every 10 seconds

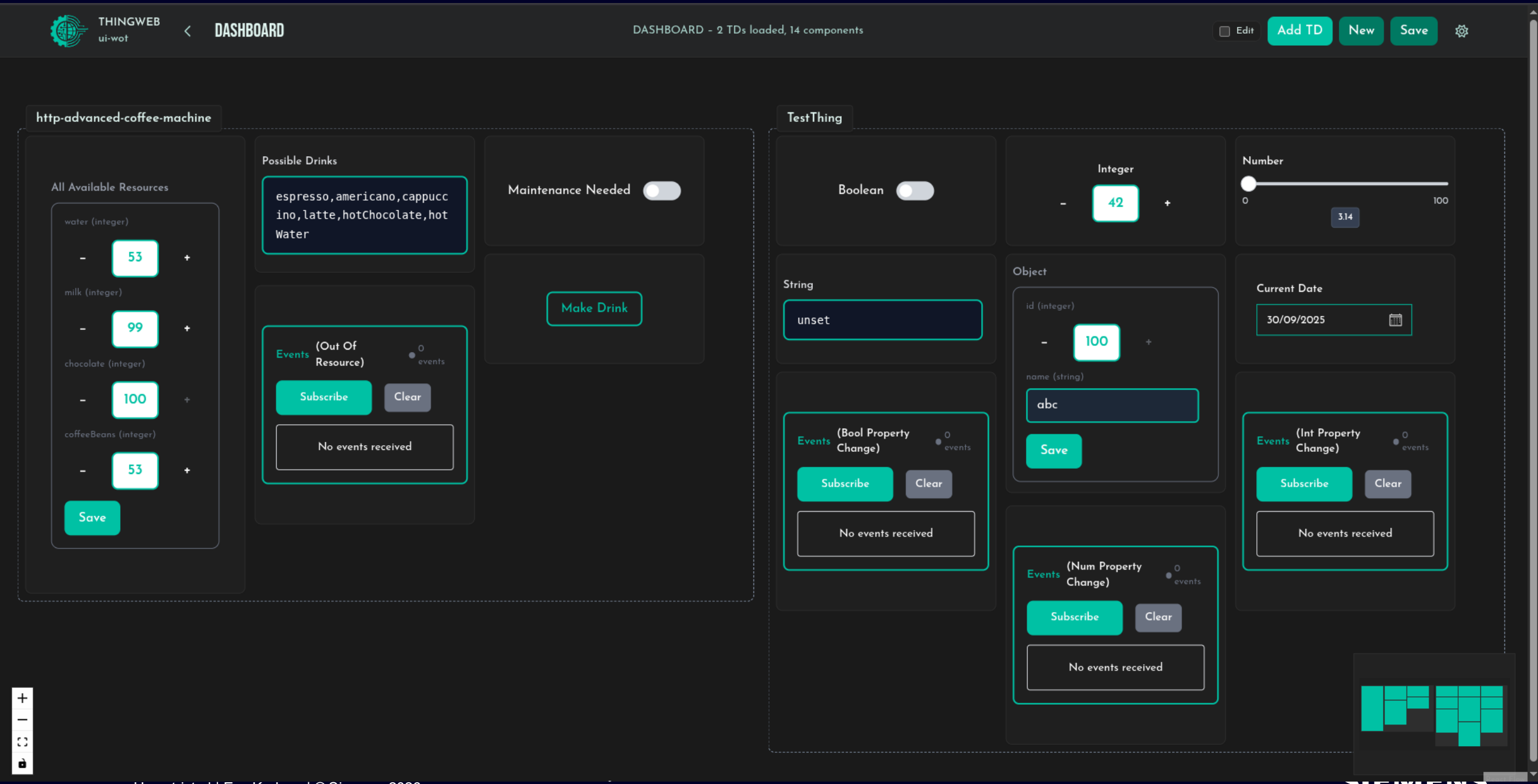
});
```

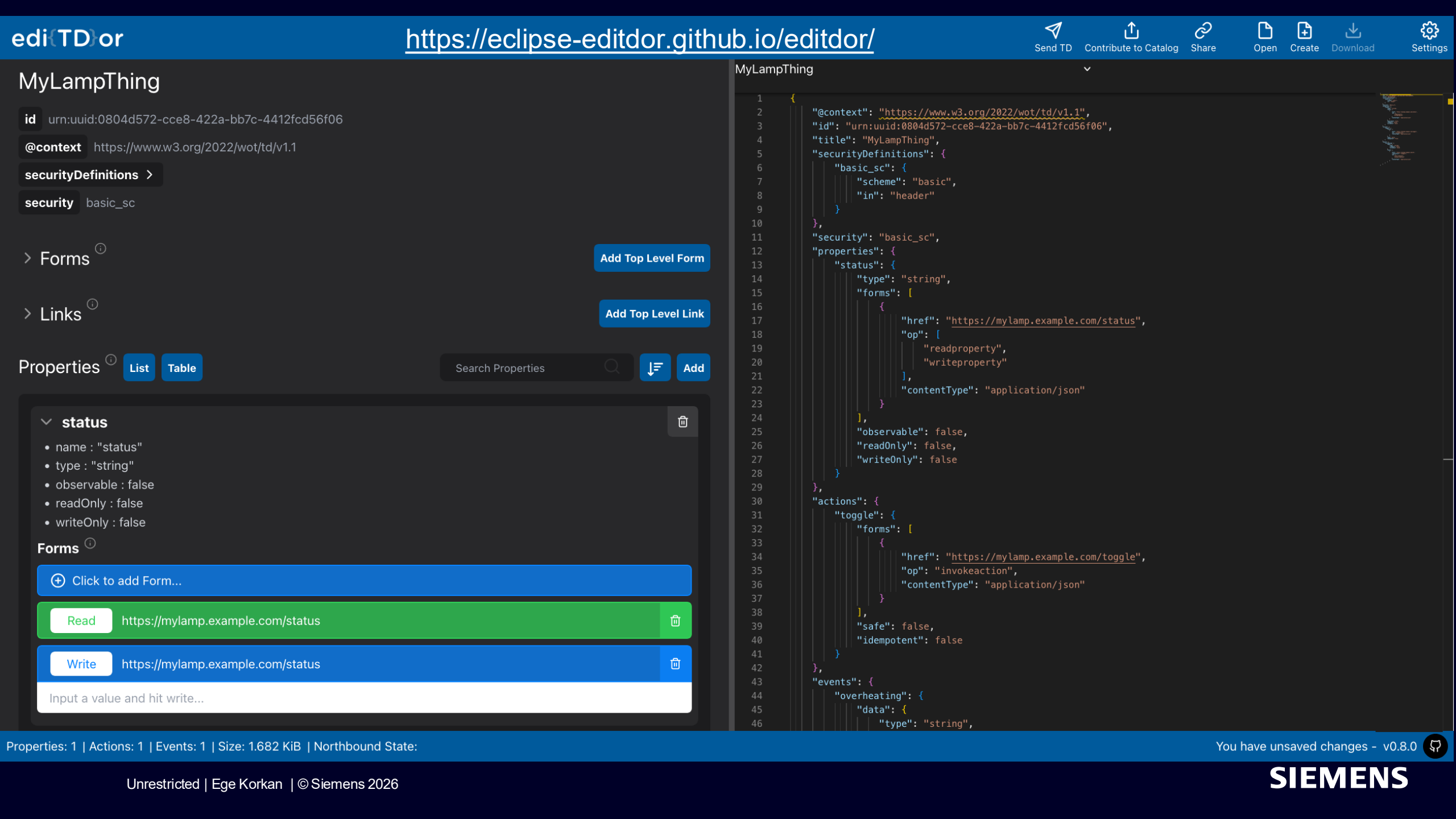
https://github.com/eclipse-thingweb/node-wot

Unrestricted | Ege Korkan | © Siemens 2026

SIEMENS

https://github.com/eclipse-thingweb/ui-wot





Unrestricted | Ege Korkan | © Siemens 2026
SIEMENS